

All the problems in this problem set are solved in some textbook or can be found on the internet. If you want to learn something then you may not want to look at these.

1. Consider the renormalization of a scalar field theory defined by the lagrangian:

$$L = \frac{1}{2} Z_2 \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} Z_m Z_2 m^2 \phi^2 - Z_1 \frac{\lambda}{4!} \phi^4,$$

where, the Z_i are the renormalization constants. By splitting the lagrangian into $L_0 + L_{int} + L_{c.t.}$ as discussed in class, determine the infinite part of the renormalization constants to one loop order. Use dimensional regularization.

2. In class we had shown that an integral of the form,

$$\int l^\mu l^\nu f(l^2) d^n l = \frac{g^{\mu\nu}}{n} \int l^2 f(l^2) d^n l.$$

Use similar arguments to argue the form of an integral like,

$$\int l^\mu l^\nu l^\alpha l^\beta f(l^2) d^n l.$$